SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:
Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

RUBRICS FOR CALCULATION

Criteria	Weightage	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Problem Analysis	15%	Clearly identifies and deeply analyzes the industry problem with all factors	Good understanding with minor gaps	Basic understanding; some key aspects missing	Poor or incorrect understanding of the problem
Solution Design	20%	Innovative, practical, and industry-relevant solution	Effective solution with minor limitations	Partially suitable solution	Inappropriate or unclear solution
Technical Implementation	20%	Fully functional, accurate, and well-executed implementation	Mostly functional with minor errors	Partially implemented solution	Incomplete or non-functional implementation
Use of Tools & Technologies	10%	Appropriate and advanced use of relevant tools and technologies	Correct use of tools	Limited or basic use of tools	Incorrect or no use of tools
Problem-Solving Approach	10%	Logical, systematic, and well-structured approach	Mostly logical approach	Somewhat disorganized approach	No clear approach
Innovation & Creativity	10%	Highly creative and original solution	Some creativity	Limited originality	No creativity
Testing & Validation	5%	Thorough testing with real-world scenarios	Adequate testing	Minimal testing	No testing
Documentation & Presentation	5%	Clear, professional, and well-documented	Good documentation	Basic documentation	Poor or no documentation
Teamwork / Collaboration	5%	Excellent coordination and contribution (if team-based)	Good collaboration	Limited participation	No collaboration or contribution

1.A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices. The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

Star Topology for a 20-Person Startup

A star topology is an appropriate network topology for a 20-person startup because it provides centralized management, easy troubleshooting, scalability, and reliable communication between devices. In this topology, all end devices are connected individually to a central network switch, which acts as the main communication point.

The network can use a 24-port managed Ethernet switch at its centre. The switch provides sufficient ports for approximately 20 employees' computers, printers, IP phones, and other network devices, while also providing additional ports for future expansion. Each workstation can be connected to the switch using Cat6 UTP cables, which support high-speed Ethernet communication and can operate over distances of up to 100 metres under standard conditions.

The central switch can be connected to a router or firewall, which provides Internet connectivity and performs important network functions such as DHCP, Network Address Translation (NAT), routing, and basic security. A wireless access point can also be connected to the switch to provide Wi-Fi connectivity for laptops, smartphones, tablets, and other wireless devices.

One of the major advantages of star topology is fault isolation. Since every device has a separate connection to the central switch, a failure in one cable or workstation affects only that particular device. The remaining devices can continue communicating normally. For example, if the network cable connected to one employee's computer becomes damaged, only that employee's connection is interrupted while the other 19 users remain connected.

Another advantage is easy network management and maintenance. The network administrator can monitor connected devices, configure ports, identify faults, and implement security policies from the central switch. Adding a new workstation is also straightforward because a new device can be connected to an available switch port without significantly affecting the existing network.

However, the main limitation of star topology is its dependence on the central switch. If the central switch fails, communication between most or all connected devices will be disrupted. Therefore, for improved reliability, the startup can use a high-quality managed switch, maintain a backup switch, and use appropriate power protection such as a UPS.

Overall, a star topology is a suitable choice for a 20-person startup because it provides centralized control, good performance, simple troubleshooting, easy expansion, and better fault isolation compared with bus or ring topologies. It offers a practical and scalable foundation for the company's wired and wireless network.

2.A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

Mesh Topology for a Banking Network

A full or partial mesh topology is highly suitable for a banking network because banks require high availability, fault tolerance, reliability, and near-zero downtime. A failure in one network link or device should not interrupt critical banking services such as online transactions, ATM services, payment processing, and inter-branch communication.

In a full mesh topology, every major network device has a direct connection to every other device. However, because a full mesh requires a large number of connections, banks generally use a partial mesh topology in the core network. In this design, important core switches, routers, firewalls, and data-centre devices have multiple redundant connections. Distribution and access switches can also be connected to at least two core switches to provide alternative communication paths.

The bank can deploy two enterprise-grade routers with redundant Internet connections. Routing protocols such as OSPF and BGP can be used to automatically select alternative routes when a link or router fails. This ensures that network traffic can continue through another available path without causing a major interruption to banking operations.

Fiber-optic cables are preferred for connections between core network devices because they provide high bandwidth, low latency, and immunity to electromagnetic interference. Single-mode fiber can be used for long-distance connections between buildings, while multimode OM4 fiber can be used for shorter high-speed connections within a data centre.

Network availability can be further improved through redundant power supplies, UPS systems, backup generators, and RAID-based storage systems. Firewalls and other critical security devices can also be deployed in high-availability pairs so that the failure of one device does not stop network services.

The major advantage of mesh topology is fault tolerance. If one network link fails, traffic can be redirected through an alternative path. Similarly, if one core device becomes unavailable, redundant devices can continue providing network connectivity. This significantly reduces downtime and helps maintain continuous banking operations.

However, a full mesh topology is expensive and complex to implement because it requires a large number of physical connections, network interfaces, and configuration resources. Therefore, a partial mesh with redundant core and distribution connections provides a better balance between reliability, cost, and manageability.

Overall, a partial mesh topology is an excellent choice for a banking network because it provides redundancy, high availability, automatic failover, fault tolerance, and reliable communication. These characteristics make it suitable for financial institutions where even a short network outage can affect customers and critical financial transactions.

3.For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.

Bus Topology for a Small Classroom Laboratory

A bus topology can be considered for a small classroom computer laboratory when the primary requirement is low installation cost and the number of computers is limited. In a bus topology, all computers share a single main communication cable, known as the backbone. This reduces the amount of cabling and eliminates the requirement for a central switch, making the initial installation relatively inexpensive.

In a traditional Ethernet bus network, 50-ohm coaxial cable such as RG-58 is used as the backbone. Each computer is connected to the backbone using a T-connector and BNC connector. A 50-ohm terminator is installed at both ends of the backbone cable to prevent signal reflection and ensure proper transmission of data signals.

The main advantage of bus topology is its low hardware and cabling cost. Since all computers use a common backbone, fewer cables and networking devices are required compared with a star topology. This makes it suitable for a small classroom where the number of computers is limited and the network traffic is relatively low.

Another advantage is its simple installation. Computers can be connected to the existing backbone without requiring a dedicated central switch. For a small laboratory used mainly for activities such as basic Internet access, file sharing, programming practice, and educational applications, the simple architecture can be sufficient.

However, bus topology has several important limitations. Since all computers share the same communication medium, network performance decreases as the number of devices and traffic increases. Data collisions can occur when multiple devices attempt to transmit simultaneously. In addition, a break or major fault in the backbone cable can disrupt communication for all connected computers, making troubleshooting more difficult.

Therefore, bus topology is suitable only for a small, low-traffic classroom environment where cost is the primary consideration. For a modern computer laboratory requiring higher performance, reliability, and easier maintenance, a star topology using an Ethernet switch would be a more appropriate choice.

Overall, bus topology provides a simple and economical networking solution for a small classroom lab, but its limitations in terms of reliability, scalability, and performance make it less suitable for larger or high-traffic networks.

4.A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

Hybrid Topology for a Large University Campus

A hybrid topology is the most suitable network design for a large university campus because thousands of users may be distributed across multiple buildings such as lecture halls, laboratories, libraries, hostels, administrative offices, and research centres. Since the requirements of each area are different, combining multiple network topologies provides better reliability, scalability, performance, and manageability.

At the campus backbone or core level, a ring or partial mesh topology can be implemented using high-speed fiber-optic connections. Core routers and multilayer switches can be interconnected using 10 Gbps or higher fiber links. The redundant connections provide alternative paths for network traffic, ensuring that a single cable or link failure does not completely disconnect a building from the campus network.

Each building can have a dedicated distribution switch connected to the campus core using dual fiber-optic uplinks. The redundant uplinks improve network availability and allow traffic to continue through an alternative connection if one link fails. The distribution layer also provides centralized control over the network within each building.

Within individual buildings, a star topology can be used. The distribution switch connects to multiple floor-level access switches, and each access switch connects individual computers, printers, IP phones, and other network devices using Cat6 UTP cables. This design makes fault detection and maintenance easier because a failure in one cable generally affects only the connected device.

For specialized areas such as computer laboratories, smaller tree or bus-based arrangements may be considered where appropriate. However, modern Ethernet networks generally prefer switched star or hierarchical designs because they provide better performance and reliability.

Wireless connectivity can also be provided throughout the campus using Wi-Fi access points connected to the access switches. Power over Ethernet (PoE) allows both network connectivity and electrical power to be supplied through a single Ethernet cable, reducing the need for separate power connections for wireless access points.

The proposed network follows a three-layer hierarchical architecture consisting of the core, distribution, and access layers. The core layer provides high-speed backbone connectivity, the distribution layer manages communication between buildings and network segments, and the access layer provides connectivity to end-user devices.

One of the major advantages of this hybrid design is scalability. When a new building is added to the university, a new distribution switch can be installed and connected to the existing campus backbone using fiber-optic links. The existing buildings and their networks do not need to be redesigned. The design also provides flexibility because different topologies can be selected according to the requirements of individual campus areas.

Overall, a hybrid topology is an ideal solution for a large university campus because it combines the redundancy of ring or mesh networks at the backbone level with the simplicity and manageability of star-based networks within buildings. It provides high availability, efficient fault isolation, easy expansion, centralized management, and reliable connectivity for thousands of users across the campus.

5.Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

Network Design for a Company with Multiple Departments

A company with multiple departments requires a structured, secure, and scalable network architecture to support communication between employees while protecting sensitive departmental information. A hierarchical network design using Layer 2 switches, Layer 3 switches, routers, VLANs, and firewalls is suitable for this type of organization.

Each department, such as Human Resources, Finance, Sales, and IT, can be connected to a dedicated Layer 2 access switch. Computers, printers, IP phones, and other devices within a department are connected to these switches. The access switches provide connectivity between devices and forward traffic to the appropriate network destination.

To logically separate the departments, Virtual Local Area Networks (VLANs) can be configured. For example, HR can be assigned to one VLAN, Finance to another, Sales to another, and IT to a separate VLAN. VLANs improve network security by isolating departmental traffic and also reduce unnecessary broadcast traffic.

The departmental Layer 2 switches are connected to a central Layer 3 switch, which performs inter-VLAN routing. This allows authorized communication between different departments while maintaining logical separation. Access-control rules can be configured to restrict sensitive communication. For example, Finance systems can be protected from unauthorized access by users in other departments.

A router or firewall can be placed at the network boundary to connect the company's internal network to external networks such as the Internet. The firewall can provide security features such as traffic filtering, access control, Network Address Translation (NAT), and protection against unauthorized network access.

For improved reliability, redundant links and backup network devices can be implemented between important network layers. If one network link fails, traffic can be redirected through an alternative path. Proper IP addressing and subnetting should also be implemented for each VLAN to organize the network efficiently and simplify administration.

The proposed design follows a hierarchical architecture consisting of access, distribution, and core functions. The access layer connects end-user devices, the distribution layer manages VLANs and inter-network communication, and the core layer provides high-speed connectivity across the organization.

Overall, this network architecture provides efficient communication, improved security, reduced broadcast traffic, fault tolerance, and scalability. It also allows new departments, users, and network devices to be added easily as the company expands, making it a practical design for a growing organization.